

Proportional Share

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#include<stdio.h>
#include <stdlib.h>
#include <time.h>
typedef struct
{
char name[5]; int tickets; }
Process;
int main() {
int n, total_tickets = 0; float total_T =0.0;
printf("1BM24CS361, Nitya R Swadi\n");
printf("Enter the number of Processes: ");
scanf("%d", &n);
Process p[n];
srand(time(NULL));
for (int i = 0; i < n; i++)
{ printf("\nProcess %d:\n", i + 1);
sprintf(p[i].name, "P%d", i + 1);
printf("Tickets: ");
scanf("%d", &p[i].tickets);
total_tickets += p[i].tickets;
total_T +=p[i].tickets;
}
printf("\n--- Proportional Share Scheduling ---\n");
printf("Enter the Time Period for scheduling: ");
int m;

scanf("%d",&m);
for (int i = 0; i < m; i++)
{
int winning_ticket = rand() % total_tickets + 1;
int accumulated_tickets = 0;
int winner_index;
for (int j = 0; j < n; j++)
{
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    accumulated_tickets += p[j].tickets;
    if (winning_ticket <= accumulated_tickets)
    {
        winner_index = j; break;
    }
}
printf("Tickets picked: %d, Winner: %s\n", winning_ticket,
p[winner_index].name);
}
for (int i = 0; i < n; i++)
{
    printf("\nThe Process: %s gets %0.2f%% of Processor Time.\n", p[i].name,
((p[i].tickets / total_T) * 100));

} return 0;
}
OUTPUT:

```

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C:\Users\Nitya\Downloads\rr X + v
1BM24CS361, Nitya R Swadi
Enter the number of Processes: 3

Process 1:
Tickets: 10

Process 2:
Tickets: 20

Process 3:
Tickets: 30

--- Proportional Share Scheduling ---
Enter the Time Period for scheduling: 5
Tickets picked: 3, Winner: P1
Tickets picked: 20, Winner: P2
Tickets picked: 54, Winner: P3
Tickets picked: 14, Winner: P2
Tickets picked: 39, Winner: P3

The Process: P1 gets 16.67% of Processor Time.

The Process: P2 gets 33.33% of Processor Time.

The Process: P3 gets 50.00% of Processor Time.

Process returned 0 (0x0)    execution time : 34.957 s
Press any key to continue.

```